Exp. No: 3

Force of Attraction between Two Electric Charges and Two Parallel Conductors Separated by Varying Distance

Aim:

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

Objectives:

1. To compute the electrostatic force of attraction or repulsion between two point charges using Coulomb's law.

2. To compute the magnetic force per unit length between two current-carrying parallel conductors.

3. To analyze and plot the variation of force with separation distance and charge or current magnitude.

Program:

Test case (i) Determination and plotting of force between two electric charges with varying distance

```
clc;

clear;

close all;


rho0 = 1e-6;      % Uniform volume charge density (C/m^3)

epsilon0 = 8.854e-12; % Permittivity of free space (F/m)

R = 1;          % Sphere radius (m)


r = linspace(0, R, 100);

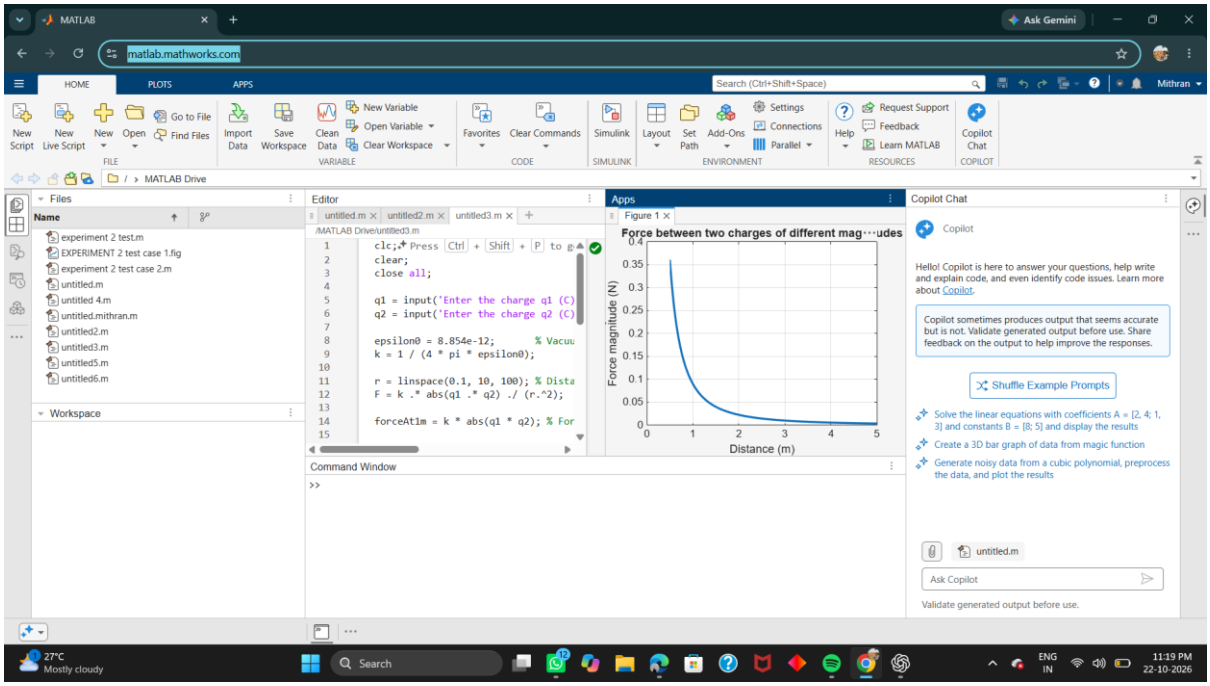

D = (rho0 .* r) ./ 3; % Electric flux density (C/m^2)

E = D ./ epsilon0;    % Electric field (V/m)


subplot(1, 2, 1);
```

```
plot(r, D, 'LineWidth', 2);
grid on;
xlabel('Radius (m)');
ylabel('Electric Flux Density D (C/m^2)');
title('Uniform Density: D vs Radius');

subplot(1, 2, 2);
plot(r, E, 'LineWidth', 2);
grid on;
xlabel('Radius (m)');
ylabel('Electric Field E (V/m)');
title('Uniform Density: E vs Radius');
```



Test case (ii) Determination and plotting of force between two electric charges with different

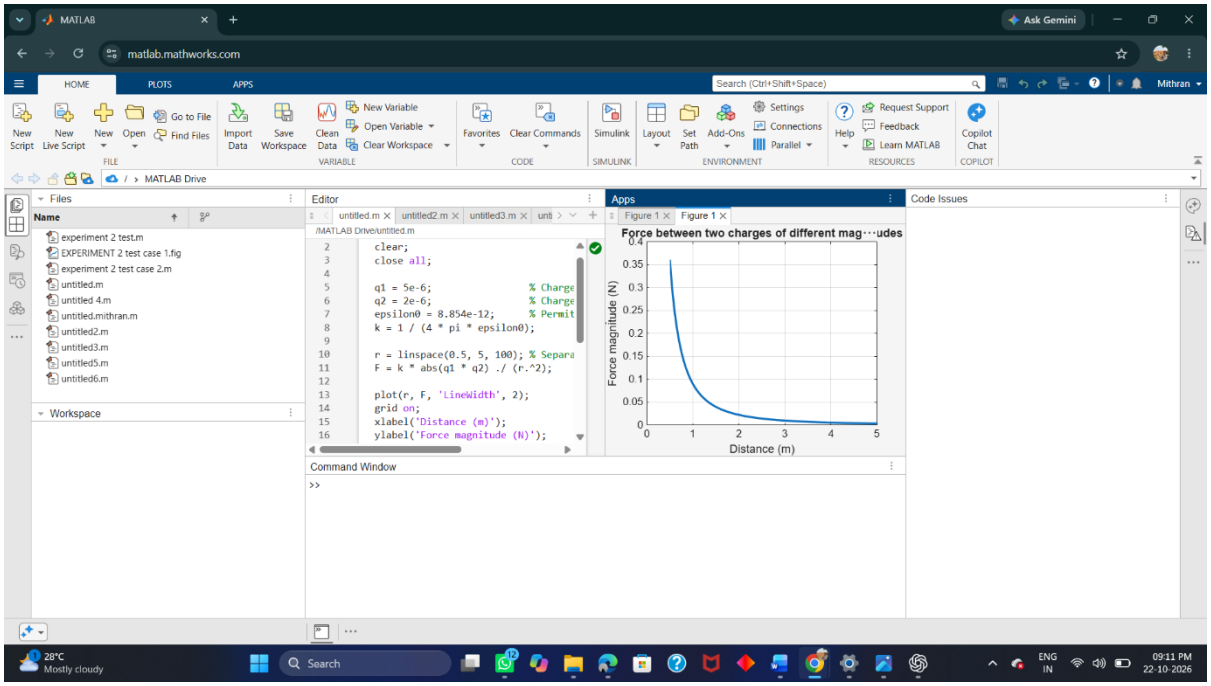
Code:

```
clc;
 q1 = 5e-6; q2 = 2e-6;
 epsilon = 8.854e-12;
```

```
k = 1/(4*pi*epsilon);
r = linspace(0.5,5,100);
F = k*q1*q2./(r.^2);
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between two charges (different magnitudes)');
```

Output:-

- Force magnitude increases with increase in charge
- Inverse square decay maintained



Test case (iii) Determination and plotting of force per unit length between two parallel conductors with varying distance

Code:

```
clc;
mu0 = 4*pi*1e-7;
I1 = 5;
I2 = 10;
```

d = linspace(0.1,10,100);

F = (mu0*I1*I2)./(2*pi*d);

plot(d,F,'LineWidth',2);

xlabel('Distance (m)');

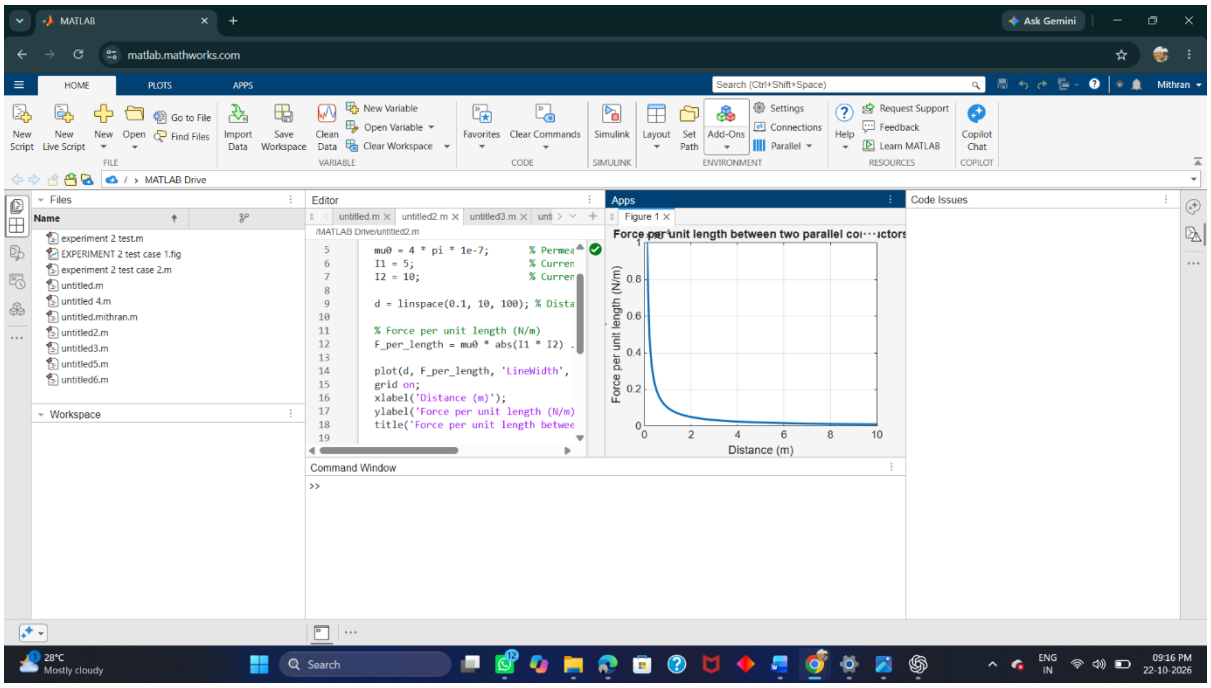
ylabel('Force per unit length (N/m)');

title('Force between two parallel conductors');

Output:

● Force decreases with increase in distance

● Linear inverse relation (1/d)



Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

Code:

clc;

mu0 = 4*pi*1e-7; I1 = 3; I2 = 3;

d = linspace(0.1,2,100);

F = (mu0*I1*I2)./(2*pi*d);

plot(d,F,'LineWidth',2);

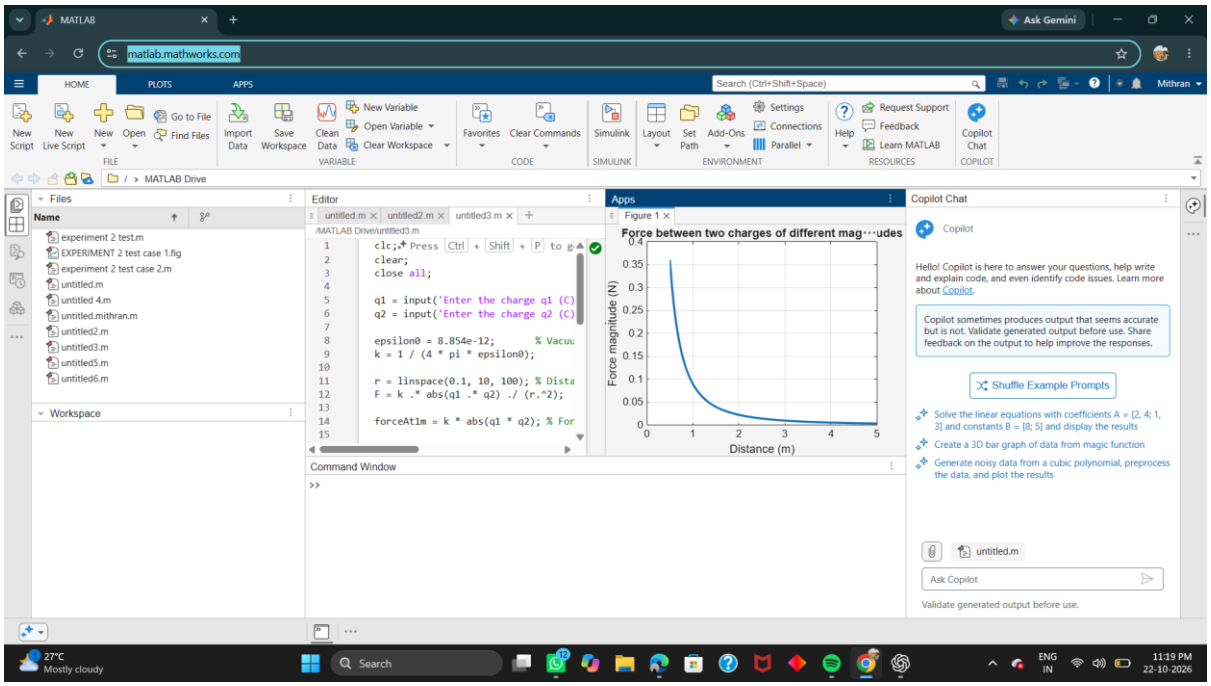
xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between conductors (equal currents)');

Output:-

Force magnitude reduced compared to Test case (iii) Same inverse relationship



Test case (v) : Magnetostatic Analysis – Biot–Savart / Ampere-Law Field Plot

MATLAB Program

clc;

clear;

close all;

%% Biot-Savart / Ampere-Law Magnetic Field Analysis

% Magnetic field around a long straight current-carrying conductor

%% Input Parameters

I = 5;   % Current in Ampere

```
mur = 1;        % Relative permeability

mu0 = 4*pi*1e-7;     % Permeability of free space (H/m)

 mu  = mu0*mur;       % Absolute permeability

 % Radial distance from conductor

 r = linspace(0.01,1,100);   % Distance in metres

 %% Magnetic Field Intensity using Ampere's Law

 H = I ./ (2*pi*r);

 %% Magnetic Flux Density B = mu .* H;

 %% Independent check: numerical Biot-Savart integration % Straight wire 100 m long (z = -50 to +50 m)

H_num = zeros(size(r));

 for k = 1:length(r)    am = atan(50/r(k));

  alpha = linspace(-am,am,20001);

   H_num(k) = I/(4*pi*r(k))*trapz(alpha,cos(alpha));

 end B_num = mu .* H_num;

%% Percentage Deviation H_error = abs((H_num - H) ./ H) * 100; B_error = abs((B_num - B) ./ B) * 100;

%% Display Selected Results fprintf('\n');

 fprintf('-----------------------------------------------\n'); fprintf(' MAGNETIC FIELD AROUND LONG STRAIGHT CONDUCTOR\n');

fprintf('-----------------------------------------------\n'); fprintf('Current I = %.2f A\n', I); fprintf('Relative permeability mur = %.2f\n', mur); fprintf('-----------------------------------------------\n'); fprintf('\n   r (m)     H (A/m)      B (T)'); fprintf('     H Error (%%)   B Error (%%)\n'); fprintf('-----------------------------------------------'); fprintf('----------------------\n');

r_selected = [0.01 0.05 0.10 0.20 0.50 1.00]; for k = 1:length(r_selected)    rr = r_selected(k);    [~,kk] = min(abs(r - rr));


  HH = I/(2*pi*r(kk));    BB = mu

*HH;    fprintf('%8.2f %10.4f %12.4e %10.4f %12.4f\n',
```

```
 r(kk), HH, BB, H_error(kk), B_error(kk)); end%% Plot H versus radial distance figure;
plot(r,H,'LineWidth',2); grid on;


 xlabel('Radial Distance r (m)'); ylabel('Magnetic Field Intensity H (A/m)');

 title('Magnetic Field Intensity H versus Radial Distance');

 %% Plot B versus radial distance figure; plot(r,B,'LineWidth',2);

 grid on; xlabel('Radial Distance r (m)');

 ylabel('Magnetic Flux Density B (T)');

 title('Magnetic Flux Density B versus Radial Distance');

 %% Verify H inverse-distance relationship figure; loglog(r,H,'LineWidth',2); grid on;

 xlabel('Radial Distance r (m)'); ylabel('Magnetic Field Intensity H (A/m)');


 title('Verification of H \propto 1/r');


%% Verify B inverse-distance relationship figure; loglog(r,B,'LineWidth',2);


 grid on; xlabel('Radial Distance r (m)'); ylabel('Magnetic Flux Density B (T)');
title('Verification of B \propto 1/r');

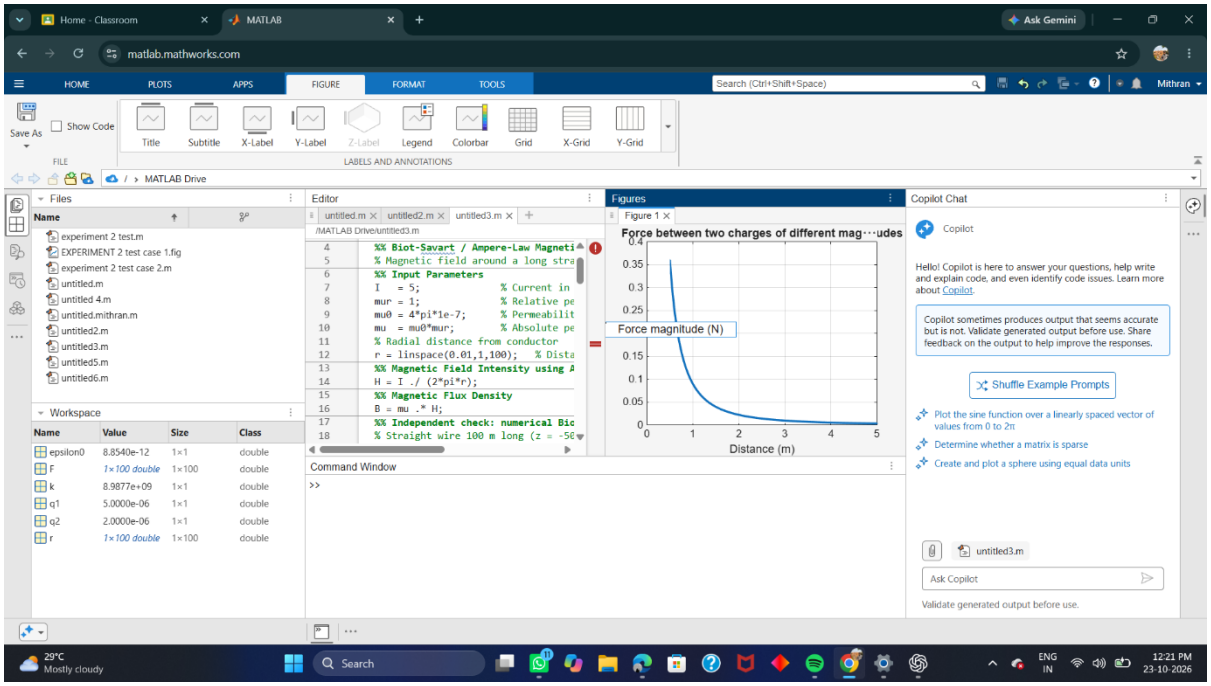

%% Calculate maximum deviation max_H_error = max(H_error); max_B_error =
max(B_error);


fprintf('\nMaximum percentage deviation in H = %.6f %%\n', max_H_error);
fprintf('Maximum percentage deviation in B = %.6f %%\n', max_B_error);
fprintf('\nRESULT:\n');


fprintf('The magnetic field intensity H and

magnetic flux density B\n'); fprintf('were calculated and plotted using MATLAB.\n');
fprintf('Both H and B decrease approximately as 1/r.\n');
```

fprintf('The MATLAB results agree with the analytical expressions.\n');



LABORATORY PERFORMANCE ASSESSMENT RUBRICS

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify physical quantitie s and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases	Identifi e s most quantities and models correctl y.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities	25	2.5	
2	Calculat e electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel current force calculations correctly with units.	Calcula t ions are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	
3	Evaluate force versus separation	Evaluate force versus separations	Correctly varies separation,	Trend and plots are correct with	. Limited sweep or incomplete	Does not establish a	30	3	

			plots trends and checks theoretical distance dependence..	minor omissi o ns.	trend interpretatio n.	valid relationship			
4	Explain and report results	Explain and report results	Explains attraction/rep ul si on and reports assumptions, units and conclusions clearly	Conclu s ion is correct with minor docum e ntation gaps..	Conclusion is incomplete or partly inaccurate	Conclusion is missing or incorrect.	20	2	
Tota l							100	10	